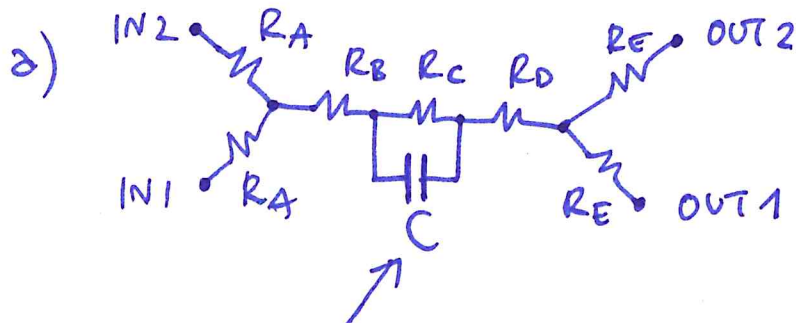


EX. 1



flux top
membrane

$$r_{eq} = \frac{2 \cdot A}{P} = \frac{2 \cdot 40 \cdot 50}{2(40+50)} = 22.2 \mu m$$

$$\gamma_A = 10^{16} \rightarrow R_A = \gamma_A \cdot A = 2 \cdot 10^{13} \left[\frac{Pa \cdot s}{m^3} \right] = R_E$$

$$\gamma_B = 2.5 \cdot 10^{15} \rightarrow R_B = R_D = 2.5 \cdot 10^{12}$$

$$R_C = 10 \cdot R_B = 2.5 \cdot 10^{13}$$

A & E



50

$$\gamma_A = \frac{7 \cdot 8 \cdot l}{\pi r_{eq}^4}$$

γ_A

C, D, B

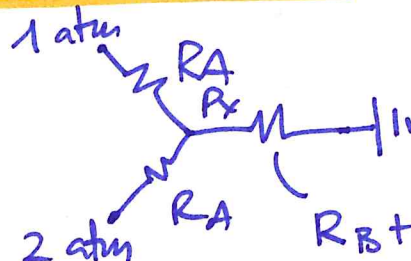
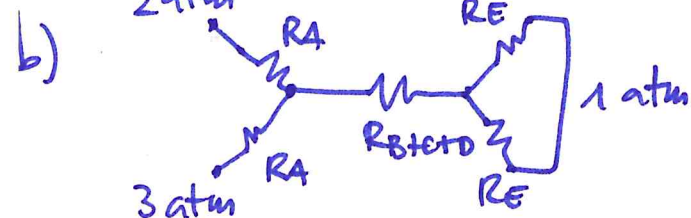


100

$$\gamma_B = \frac{12 \cdot 7 \cdot l}{(1 - 0.63 \frac{w}{h}) h^3 \cdot w}$$

γ_B

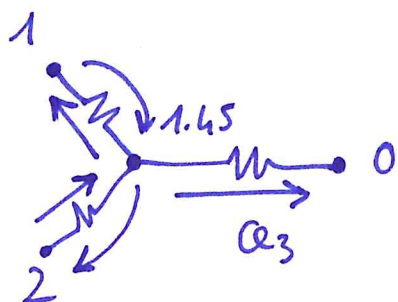
1/4



SUPERPOSITION :

$$P_X = 1 \text{ atm} \cdot \frac{R_{TOT} // R_A}{R_{TOT} // R_A + R_A} + 2 \text{ atm} + ()$$

$$= 3 \text{ atm} \cdot \frac{40 // 20 \cdot 10^9 (13.3)}{40 // 20 \cdot 10^9 + 20 \cdot 10^9} = 1.2 \text{ atm}$$



$$Q_1 = - \frac{(1.45^2 - 1) \cdot 101 \text{ kPa}}{R_A} = - \frac{1}{20 \cdot 10^9} \cdot 10^{-9} \frac{m^3}{s}$$

$$Q_2 = \frac{(2 - 1.45^2) \cdot 101 \text{ kPa}}{R_A} = \frac{4}{20 \cdot 10^9} \cdot 10^{-9} \frac{m^3}{s}$$

$$Q_3 = Q_2 - |Q_1| = \frac{1.45^2}{R_{TOT}} = 3 \cdot 10^{-9} \frac{m^3}{s}$$

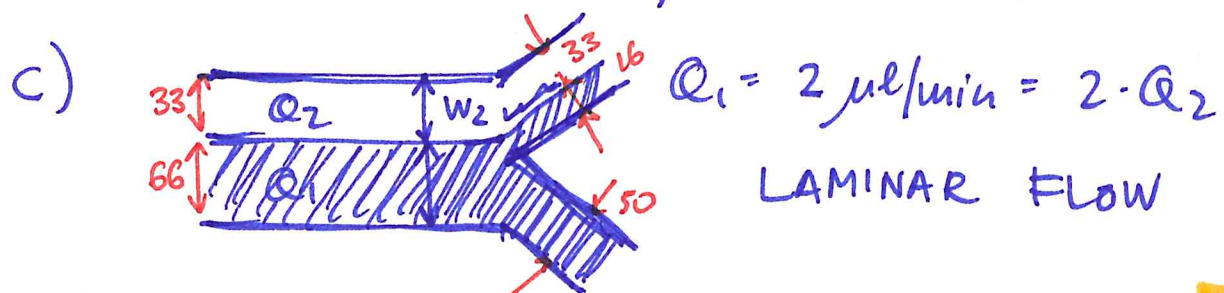
$$V_{A1} = - \frac{1 \cdot 10^{-9} \text{ m}^3}{5} \cdot \frac{1}{h \cdot w_a} = - \boxed{0.5 \text{ m/s}}$$

$$V_{A2} = 4 \cdot |V_{A1}| = \boxed{2 \text{ m/s}}$$

$$V_{B,C,D} = \frac{Q_3}{h \cdot w_b} = \frac{3 \cdot 10^{-9} \text{ m}^3/\text{s}}{40 \mu\text{m} \cdot 100 \mu\text{m}} = \boxed{0.75 \text{ m/s}}$$

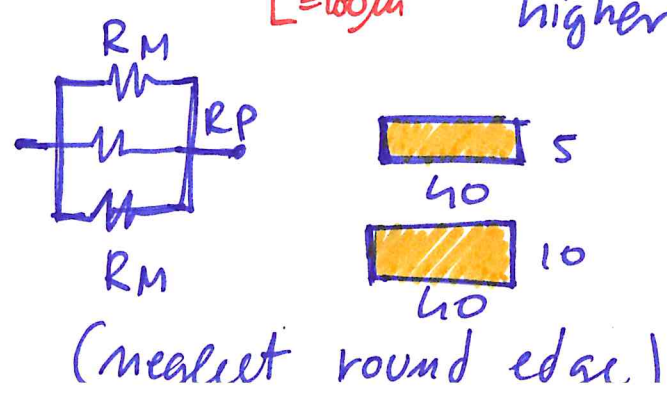
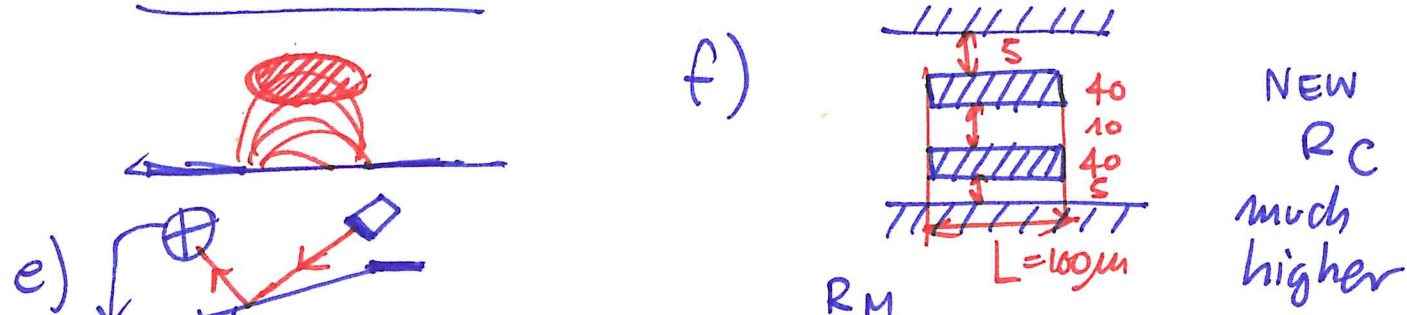
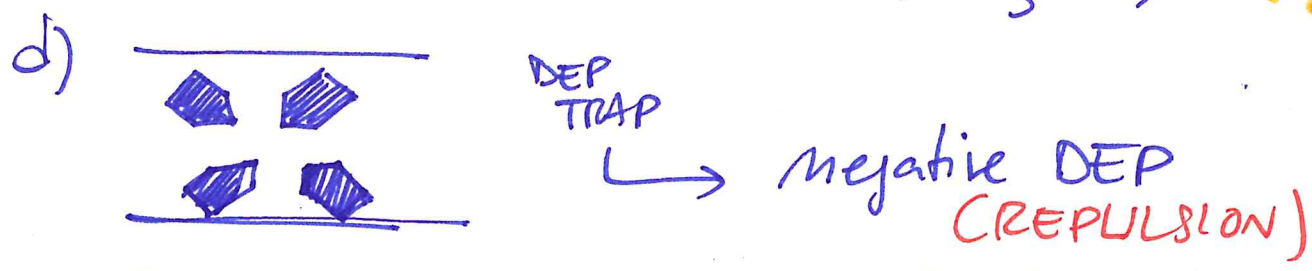
$$V_E = \frac{Q_3/2}{h \cdot w_a} = \frac{3/2 \cdot 10^{-9}}{40 \mu\text{m} \cdot 50 \mu\text{m}} = \boxed{0.75 \text{ m/s}}$$

GIVEN THE SYMMETRY, IF SWAPPED → SIGNS ARE REVERSED



$$w_1 = 2w_2 \quad \begin{cases} w_1 + w_2 = 100 \mu\text{m} \\ w_1 = 2w_2 \end{cases} \quad w_2 = \frac{100 \mu\text{m}}{3} = \boxed{33 \mu\text{m}}$$

$$w_1 = \frac{2}{3} \cdot 100 \mu\text{m} = \boxed{66 \mu\text{m}}$$



$$R_M = \frac{12 \cdot \eta \cdot L}{(1 - 0.63 \frac{s}{40}) (5 \mu m)^3 \cdot 40 \mu m} = 2.4 \cdot 10^{14} \left[\frac{Pa \cdot s}{m^3} \right]$$

$$R_P = \frac{12 \cdot \eta \cdot L}{(10 \mu m)^3 \cdot 40 \mu m} = 3 \cdot 10^{13} \left[\frac{Pa \cdot s}{m^3} \right]$$

$$R = \frac{R_M}{2} \parallel R_P \sim 2.4 \cdot 10^{13}$$

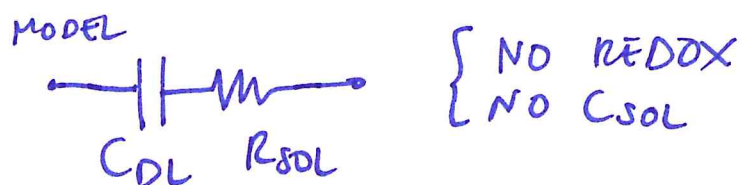
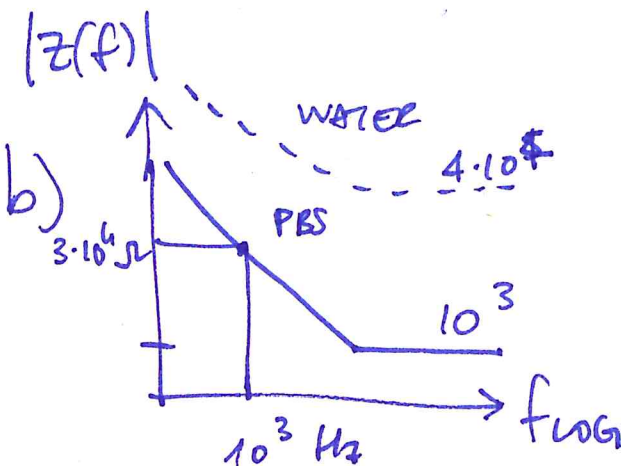
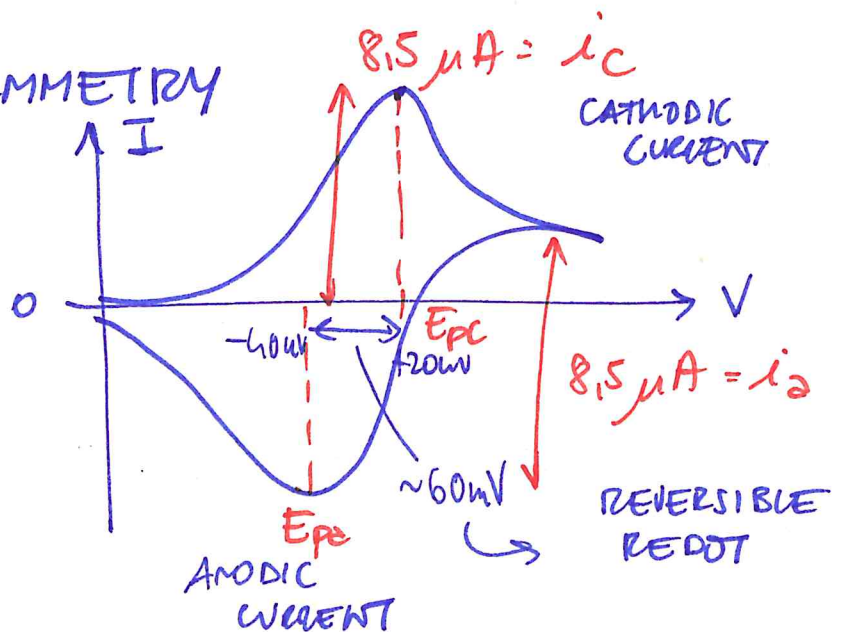
	DEP TRAP	CUP TRAP
PRO	NO CLOGGING CONTROLLABLE	PASSIVE
CON	NEED TUNE FREQ. CLAUSIUS MOSSOTTI	RISK CLOGGING HIGHER RESISTANCE
	NEED ELECTRODES	SINGLE USE

EX. 2

2) (A) CYCLIC VOLTAMMETRY

SCAN RATE $\begin{cases} 10 \text{ V/s} \\ 1 \text{ V/s} \end{cases}$

$i_p \sim \sqrt{SR}$ confirmed from plot



$$R_{SOL} = 10^3 \Omega = p.k$$

$$C_{DL} = \frac{1}{2\pi f |Z|} = \frac{1}{2\pi \cdot 10^3 \cdot 3 \cdot 10^4} = 5.3 \text{ nF}$$

Area electrode: $\frac{5.3 \mu F}{0.1 pF/\mu m^2} = 53,000 \mu m^2$

$\sqrt{53,000 \mu m^2} = 230 \mu m$ SIDE

$R \propto \rho$ $\frac{10^3 \Omega}{1/1.5 S/m} = \frac{4 \cdot 10^4}{1/\sigma}$

$10^3 \cdot 1.5 = 4 \cdot 10^5 \cdot \sigma$

$\sigma_{WATER} = \frac{10^3 \cdot 1.5}{4 \cdot 10^4} \approx 0.0375 S/m$

c)

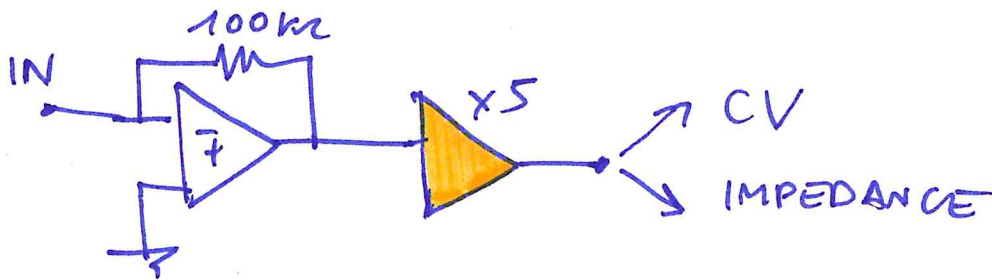
	(A) CV	(B) IMPEDANCE
I_{max}	$8.5 \mu A$	VAC / R_{MIN}
BW	$40 \cdot SR_{max}$ 400 Hz	<u>500 kHz</u>

$\frac{10 mV}{1 m\Omega} = 10 \mu A$

MORE STRINGENT

$R_F = \frac{5V}{10 \mu A} = 500 k\Omega$ chn BW = $\frac{1}{2\pi \cdot 0.2 pF \cdot 500 k\Omega} \approx 1.6 MHz$ TIGHT

REDUCE R_F to 100 k Ω $\rightarrow$ BW = 8 MHz



d) $\frac{4 \mu T}{R_F} \dots$

e) $BW|_{u4} = 10 \cdot \frac{1}{T} \sim 10 kHz$ $f|_{mv} > 20 MHz$

